SCHEDULE 3:BANK CREDIT ANALYSIS (FINNACIAL STATEMENT ANALYSIS, ETC.)

FINANCIAL STATEMENT ANALYSIS

A bank customer may wish to collect a *loan* or *credit facility*. He will have to present a proposal that will show the total amount he wishes to source from the bank, his own contribution and how he will go about the expenditure involved so as repay duly, etc.

The above is the summary of what a proposal is. Besides the expenditure and loan facilities, there abound a number of other inputs that the customer and the bank will agree upon. One of such fundamental inputs is the *Financial Statement* of the prospective borrowing customer that contains:

1) Profit & loss and Balance sheet plus appropriation account
2) Cash flow statement
3) Statement of fund application e.t.c.

One major benefit of financial statement is the use of the first item above (1)…i.e. P&L and Balance Sheet to make calculation of certain ratios. The essence is to estimate the financial capability or otherwise of the organization.

*Recall that FINANCIAL STATEMENT ANALYSIS or RATIO ANALYSIS was taken under **FIN 210** for the sole purpose of *calculation*; in **FIN 320** for *calculation* and *interpretation*. However in **FIN 417** it is meant for *calculation, interpretation* and *application* for assessment of loan proposal.

DEFINITION

Ratio Analysis may be defined as the information obtained in P&L and Balance Sheet (including appropriation account) for the purpose of assessing the financial strength (or weakness) of a firm.

OBJECTIVE

Broadly speaking, the main purpose of ratio analysis is to make *comparisons* in three main ways viz:

(1) Inter firm comparison (2) Inter temporal (3) Comparison between a firm and the industry where a firm operates.

1) **INTER-FIRM COMPARISON/PEER GROUP ANALYSIS:** These are comparisons that are made between two or more firms operating in the same industry. E.g. UBA PLc. and Union Bank PLc; both operate in the banking industry.
2) **INTER-TEMPORAL/TREND ANALYSIS:** These are comparisons that could be made within a single firm or company over two or more periods e.g. MTN 2019 performance compared to that of 2018.
3) **COMPARISON BETWEEN A FIRM AND INDUSTRY's AVERAGE:**
   e.g. measuring the performance of UBA Vs. Average performance of the Nigerian banking industry.

   Illustration 1:

   Find the average growth rate of Nigerian banking industry if the following figures were posted (assuming only 5-banks exist).

Selected Banks	Profit Growth Rate
FBN	15%
WEMA	10%
GTB	25%
FBN	20%
ZENITH	30%

Av. Growth Rate: 100 ÷ 5 = 20%

FORMS/TYPES OF RATIOS

1. **Short-term Solvency Ratio**

   i) Current ratio $CA/_{CL}$

   ii) Acid test ratio/Quick ratio $= \frac{CA - Stock}{CL}$

   iii) Net working capital = $CA - CL$

2. **Activity Ratio**

   i) Average collection period/Debtors days $= \frac{Trade\ debtors}{credit\ sales}$ x 365 days

   ii) Credit payment period/Creditors days $= \frac{Trade\ creditors}{credit\ purchases}$ x 365 days

   iii) Stock turnover $= \frac{Average\ stock\ (Op,+cl)\ \ /_2}{Cost\ of\ sales\ \ (COGS)}$

   iv) Debtors turnover $= \frac{Sales}{Debtors}$ %

v) Total asset turnover = $\frac{Sales}{Total\ assets\ (tangible\ assets)}$

vi) Fixed asset turnover = $\frac{Sales}{Fixed\ tangible\ assets}$

vii) Capital employed turnover = $\frac{Sales}{Total\ CE}$

3. **Profitability Ratio**

i) ROSA = $\frac{NPBIT}{sales}$ x 100%

ii) ROTA = $\frac{NPBIT}{Total\ tangible\ asset}$ x 100%

iii) ROCE = $\frac{NPBIT}{Total\ long\ term\ capital}$ x 100%

iv) ROSE = $\frac{PATOSH}{Ordinary\ share\ holders'\ Equity}$ x 100%

4. **Long-Term Solvency Ratios/Capital Structure Ratio**

i) Proprietary ratio = $\frac{Ordinary\ shares+Reserves+Preference\ shares}{Total\ assets\ or\ Total\ liabilities}$

ii) Debt ratio = $\frac{Long\ term\ debt+\ Short\ term\ debt}{Total\ assets\ or\ Total\ liabilities}$

iii) Debt – equity ratio = $\frac{Long\ term\ debt+\ Preference\ shares}{Ordinary\ shares+Reserves}$ x 100%

v) Fixed interest cover = $\frac{NPBIT}{Interest\ on\ debt}$

5. **Capital Market Ratio**

i) EPS = $\frac{PATOSH}{No.of\ Ord.\ shares}$

ii) DPS = $\frac{Total\ Div.\ Declared}{No.of\ Ord.\ shares}$

vi) Earnings yield = $\frac{EPS}{MPPS}$

vii) Price-earnings ratio $\frac{MPPS}{EPS}$

KEY:-

PATOSH	Profit Available to Ord. Sh./Holder
ROSA	RETURN on Sales
ROTA	RETURN on Total Assets
ROCE	RETURN on Capital Employed
ROSE	RETURN on Share Holders' Equity
EPS	Earnings Per Share
DPS	Dividend Per Share
MPPS	Market Price Per Share

INTERPRETATIONS

LIQUIDITY RATIOS:

1) **Current ratio:** The purpose of this ratio is to find out if a firm or a company can adequately meet it current obligations as at well due.

2) **Acid test ratio:** This ratio measures "how more quickly" a company can meet its current obligation.
3) **Networking capital:** It simply expresses the net position between current asset and current liabilities in Naira terms.

ACTIVITY RATIOS:

1) **Debtors' days:** This ratio measures the average number of days per year in which the company sales are tied-up in trade debtors.
2) **Creditors' days:** This ratio measures the average number of days per year in which a customer owes his supplier of trade goods.
3) **Stock turnover:** It measures the impact of average of stock recorded on the cost of inventory or cost of goods sold.
4) **Debtors' turnover:** This ratio measures the number of times the company's credit sale covers its debtors.
5) **Total asset turnover:** Measures the number of times the company credit sales can cover its total assets.
6) **Fixed asset turnover:** Measures the number of times company's sales can cover its fixed asset.
7) **Capital employed turnover:**

PROFITABILITY RATIO:

1) **ROSA:** This ratio indicates the percentage of sales that are generated as profit.
2) **ROTA:** Indicates the percentage of total assets that are generated as profit.
3) **ROCE:** This ratio indicates the percentage of used long term capital that are generated as profit.

4) **ROSE:** It measures the amount/percentage of the company's ordinary shareholders Equity (i.e. ordinary shares + reserves) that are generated as profit.

LONG-TERM SOLVENCY RATIO:

1) **Proprietary ratio:** It measures shareholders/owners' proportion of the total long term capital contribution to the company. That is internal contribution.
2) **Debt ratio:** It measures the portion of capital contribution by outsiders. i.e. borrowed capital from lenders.
3) **Fixed interest cover:** It calculate the number of times a company's operating profit can repay interest on borrowed money or debt.

CAPITAL MARKET RATIO:

1) **EPS:** It reveals how much you are earning on every share you own.
2) **DPS:** The dividend collected from ordinary shares owned by shareholders.
3) **Earning yield:** This refers to your earnings from every current market value of your shares in a company.
4) **Dividend yield:** Refers to the dividend collected from every current market value of your shares.

KEY BALANCE SHEET ITEMS'ANALYSIS FOR BANK LENDING PURPOSE

For sake of convenience, we can divide the balance sheet into its two (2) traditional parts (i.e. assets and liabilities). The assets side can further be subdivided into fixed and current assets. Also the liability side can be subdivided into long term liability and current liability. We shall consder them in turn.

FIXED ASSETS

1) **Land & building:** By land we refer to a fixed immovable asset that can either be plain land (undeveloped) and developed.

   IN ORDER TO ACCEPT A FIXED ASSET AS COLLATERAL FOR LOAN, BANK SHOULD:

   a. Investigate if it is loca ted in an acceptable area for the sales of collateral security. The advantages include ease of sale and higher value.
   b. Derive the current market value of the land.
   c. Find out the present use of the land.
   d. Ensure that all the necessary legal documentation are in place e.g. C of O.

2) **Plant & machinery**
   a. Find if it is obsolete
   b. Find if it can be upgraded or rejected for a new purchase
   c. Find the current market price if it
   d. Find if there is a restriction on its use
   e. Find its capacity utilization
3) **Furniture & fittings** (Similar consideration as in 2 above.)

CURRENT ASSETS

The common current assets are discussed as follows:

STOCK:

Where we recall that balance sheets of corporate entities are prepared in the order of illiquidity; this will help to remind us that "stock" is the slowest moving items amongst all the items under current assets! This means stock-in-trade i.e. what the bank customer sells (e.g. spare parts) is the slowest when it comes to converting to cash. This is because after sale on credit, you have to convert to debtors before you are paid in Cheque (i.e. cash at bank) or cash – the utmost liquidity.

Note: if liquidity means "nearness to cash", then cash itself is the ultimate liquidity.

Stock figures are very crucial in every trading business because it has a lot of implications. These are further broken down as follows:

A, WHERE A FIRM CARRIES TOO MUCH STOCK TO SELL :

a) such company will incur CARRYING STOCK; i.e. the opportunity cost of utilizing the money tied down in stock. E.g. loss of interest rate on savings.

b). cost of deteriorating stock kept idle. This can be enormous atimes

c) cost of leakage assuming a gas-selling company

d) cost of storage or space occupied by unutilized stock

e) cost of obsolescence for outdated stocks

d) cost of pilferage or theft.

e) increased cost of employing extra store keeper

f) loss incurred as the stock may be sold at discount to offload filled store.

A, WHERE A FIRM CARRIES TOO LITTLE STOCK :

a) such company will incur untold STOCK-OUT COSTS that are innumerable and includes:-

b). cost of production outage as the plant will seize to produce.

c) some machines might break down where production stops abruptly

d) loss of customers' confidence where production and supply fails

e) the firm may be forced to buy at higher price to keep customers

f) idleness may force stoppage of operation, retrenchment, etc.

g) a total stock-out for some regularly active producing company like UAC (producer of "Gala"–a relatively high consumable item) can be quite enormous.

DEBTORS:

This is the product of credit sales. Normally, a firm will sell on credit or discount to attract customers in a competitive business environment where winning and keeping your customers matter a lot.

Inspite of the rationale for maintaining a reasonable level of debtors, it is not advisable to embark on explosive accumulation of debtors. The issues pertaining to carrying debtors on a firm's balance sheet thus beg some salient questions like:

a)how many days are tied up in debtors?

b) how chronic is their indebtedness?

c) can they be motivated to pay up if some discounts are given?

d) can the firm employ the service of debt recovery agents?

e) what has been the firm's experience regarding bad debt?

BANK BALANCE AND CASH AT HAND

Cash, either at bank or at hand represents liquidity. It is good for a firm to be liquid so as to meet current obligation as they fall due. In equal vein, it is advisable for an efficient firm not to maintain a cash balance that it too large. This is due to some reasons as follows:

a) idle cash indicate loss of return on investment.

b) risk of theft is higher with a firm maintaining too much cash

c) holding huge cash balance may prompt/trigger wasteful spendings

LONG-TERM LIABILITY/CAPITAL

Long term liabilities generally represent the long term obligations that keep the operating firm surviving not only in the present but also long term to come. These are long term investments made by equity holders and debt holders and can also be seen as long term capital contributions.

The most common of these elements are:

-ordinary shares

-preference shares

-debentures or long term loan

-reserves which covers retained earnings and other reserves

The four (4) above can broadly be reffered to as equity and debt capital contributions by stake holders. We can further explain them in turn as follows;

WHERE A COMPANY UTILIZES MORE EQUITY THAN DEBT

a) more capital can be raised through equity than debt

b) more use of equity capital allows for recruitment of good board members

c) use of equity than debt negates the risk of liquidation

d) equity holders are owners of the firm, hence less pressure on compulsory dividend payments

e) where a firm is less geared (uses lesser debt than equity), it can further borrow conveniently when need arises

WHERE A COMPANY UTILIZES MORE DEBT THAN EQUITY

a) it can enjoy "tax-deductibility" advantage of interest rate

b) debt instruments are easier to manage than equity

c) use of debt capital calls for more discipline for fear of liquidation

d) raising debt capital is far less expensve than equity capital

e) some debt instruments can be converted to equity based on agreement

CURRENT LIABILITY

The contents of current liabilities are mainly:

-TRADE CREDITORS

These are suppliers of trade goods. For a firm trading in spare parts, for instance, its stock-in-trade are spare parts while the suppliers of the spare parts are the trade creditors.

It is good to maintain a stable supply of goods to avoid such unwanted outcomes as stock out for a borrowing company, it is good it settles suppliers obligation as when due.

-BANK OVERDRAFT

This is a sort of 'permission' accorded bank customer to overdraw his account up to a certain limit.

It should be noted that it is one major way to test the integrity of a bank customer.

-ACCRUED EXPENSES

These are expenditures incurred but has not been settled or paid for by a business firm. Sometimes such could be enormous and count significantly towards a company's short term debt. This development should give concern to a lending banker since they must at one point in time or another be paid their entitlement - for instance, salaries and wages of firm employees. 'A labourer deserves his wage' so goes the saying. The issue at this point is when the payments are made, by how much will it affect the chance of settling banks' indebtedness?